

5.7 Classwork: Introduction to Kinematics

Key definitions. A *particle* moves along a straight line. Its **position** s is measured from a fixed origin, with positive to the right.

Displacement — change in position

Distance — total length of path

Velocity — rate of change of displacement

Speed — magnitude of velocity

Acceleration — rate of change of velocity

vector (has sign: direction matters)

scalar (always ≥ 0)

$$v = \frac{\Delta s}{\Delta t} \quad (\text{vector})$$

$$|v| \quad (\text{scalar, always } \geq 0)$$

$$a = \frac{\Delta v}{\Delta t} \quad (\text{vector})$$

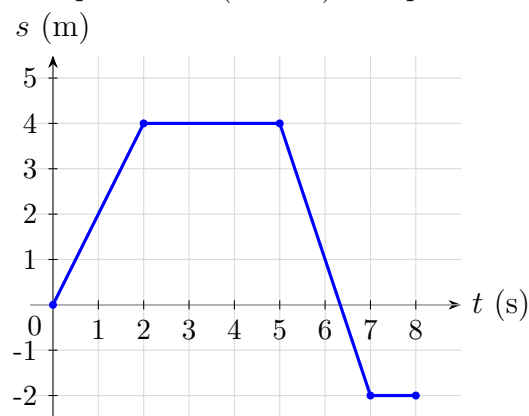
1. A runner jogs 400 m east and then 150 m west.

(a) Find the total **distance** the runner travels. [1]

(b) Find the runner's **displacement** from the starting point. State the direction. [2]

(c) Explain why these two answers are different. [1]

2. The graph below shows the position s (metres) of a particle at time t (seconds).



(a) Find the velocity of the particle during $0 \leq t \leq 2$. [2]

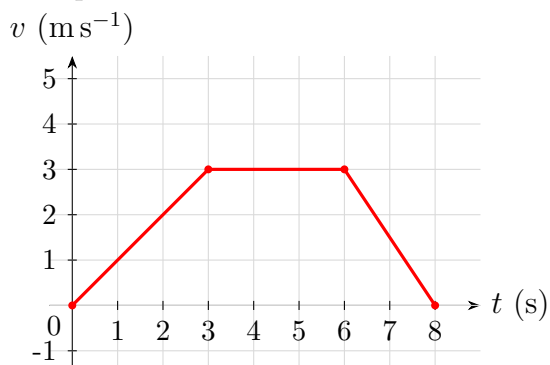
(b) Describe the motion of the particle during $2 \leq t \leq 5$. [1]

(c) Find the velocity during $5 \leq t \leq 7$. [2]

(d) Find the total **distance** travelled from $t = 0$ to $t = 7$. [2]

(e) Find the **displacement** from $t = 0$ to $t = 7$. [1]

3. The velocity v (m s^{-1}) of a particle is shown below.



- (a) Find the acceleration during $0 \leq t \leq 3$. [2]
- (b) State the acceleration during $3 \leq t \leq 6$. [1]
- (c) Find the acceleration during $6 \leq t \leq 8$. [2]
- (d) Is the particle slowing down during $6 \leq t \leq 8$? Explain your answer. [1]
4. For each quantity, state whether it is a **scalar** or a **vector**. [4]
- (a) A car travels at 60 km/h. (c) Velocity $v = -5 \text{ m s}^{-1}$.
- (b) A car drives 3 km north. (d) Temperature is 22°C .
5. A particle moves along a line. At time $t = 4 \text{ s}$, its velocity is $v = -3 \text{ m s}^{-1}$.
- (a) What does the negative sign tell you about the particle's motion? [1]
- (b) Find the speed of the particle at $t = 4$. [1]
6. The table shows the position of a particle at different times.
- | | | | | | |
|---------|---|---|---|----|----|
| t (s) | 0 | 1 | 2 | 3 | 4 |
| s (m) | 2 | 5 | 8 | 11 | 14 |
- (a) Find the velocity of the particle. [2]
- (b) Find the acceleration of the particle. [1]
- (c) Describe the motion in words. [1]
7. A cyclist starts from rest and accelerates uniformly at 2 m s^{-2} for 5 seconds.
- (a) Find the velocity of the cyclist at $t = 5 \text{ s}$. [2]
- (b) Sketch a velocity–time graph for $0 \leq t \leq 5$. [2]